



MarketLink

Project Report

TechWiz 2026 · End-to-End Web Solutions

Team Omniverse · Aptech Learning Centre, F.B. Area, Karachi

Note for the team (please read before submitting). The SRS (section "Important Note Regarding AI Usage") does not allow documentation that is fully produced by an AI tool. This report is a **draft** that Claude Code (an AI coding assistant) prepared from the project's code, database and test runs so that the facts, tables and diagrams are correct. Before you submit it, read every section, rewrite the text in your own words, add your own design reasoning and team details, and keep "Claude Code (Anthropic)" in the list of AI tools you used (section 11). The report contains no source code, as the SRS asks.

1. Problem definition

Local farmers markets in Pakistani cities (for example the weekend markets in Clifton, DHA or F-9 Park) work only face to face. That causes the same problems every week:

- **Customers do not know what is there before they go.** They cannot see which farmers come on which day, what they have picked this week, the price, or whether it is already sold out.
- **Farmers guess how much to bring.** Without orders in advance they either bring too much (waste) or too little (lost sales), and they cannot tell regular customers that they will not come.
- **Popular produce sells out early** and people who arrive later leave with nothing.
- **Nobody manages the markets online.** There is no single place for market days and hours, the stall locations, trusted reviews or complaints.

MarketLink puts the farmers market online without changing how it works: farmers publish their weekly stock, prices and pickup windows; customers find markets and farmers on a map, reserve products with a pre-order for a pickup time slot, and **pay the farmer in person at pickup**. An administrator approves farmers, manages markets and categories, moderates content and reads platform reports.

1.1 Objectives

1. Let customers browse markets, farmers and products by location, day, category and price, with a map and directions to the pickup point.
2. Let customers pre-order against the farmer's real stock for a pickup slot, then track, modify or cancel the order before the farmer's cut-off time.
3. Give farmers a simple dashboard for weekly stock (with a recurring template), pickup windows, pre-orders, sales insights and replies to reviews.
4. Give the administrator control over users, markets, categories, content, announcements and reports.
5. Work well on phones, in English and in Urdu, with secure, role-based access.

1.2 Scope

In scope	Out of scope (SRS 1.5)
Customer, farmer and admin areas with role-based access	Online payment (payment is made at pickup)
Markets, farmers, products, categories, cities	Delivery or courier logistics (pickup only)
Pre-orders with pickup slots and cut-off times	Checking farmer identity, licences or organic certificates
Reviews, ratings, favourites, restock reminders	Native mobile apps (the website is responsive instead)
Maps (OpenStreetMap), directions, notifications and e-mail	
AI assistant (English, Urdu, Roman Urdu)	

1.3 Users

User	What they do
Guest	Browses markets, farmers and products, uses the map and the assistant, can check out as a guest (an account is made and the password e-mailed)
Customer	Everything a guest can, plus a dashboard, order tracking, changes and cancellations, favourites, family sharing, reviews, restock reminders and notifications
Farmer	Registers a stall (approved by the admin), manages products, weekly stock, pickup windows, closed dates, pre-orders, sales insights and review replies
Administrator	Separate dashboard: farmers, customers, markets, cities, categories, products, reviews, moderation, announcements, the home page offer banner, messages, newsletter, FAQs, reports and farmer rankings

2. Design specifications

2.1 Architecture

MarketLink is a **MERN** application (MongoDB, Express, React, Node.js) with the multi-tier structure the SRS describes: the browser runs the React app, the Express server exposes a REST API and serves the built app, and MongoDB stores the data.

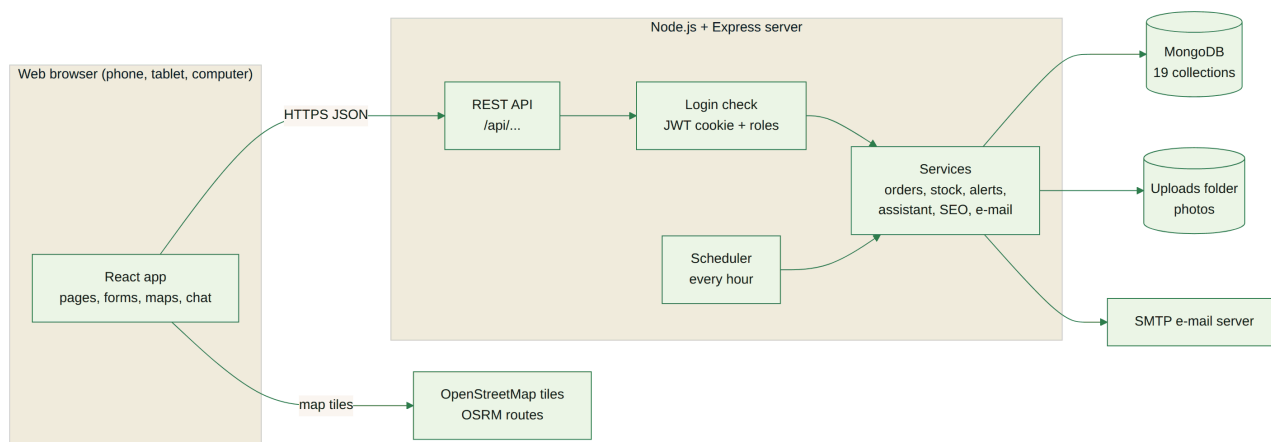


Figure 1 – System architecture

Layer	Technology	Notes
Presentation	React 19 (Vite), React Router 7, Bootstrap 5 with a custom theme, Bootstrap Icons, Recharts, DataTables, Leaflet maps, intl-tel-input	77 routes (public, customer, farmer, admin); English and Urdu (right to left)
Application	Node.js 20+, Express 5	146 REST endpoints under <code>/api</code> , input checks on every request, rate limits on login, chat and AI features
Security	JWT in an httpOnly cookie, bcrypt password hashes, Helmet headers, CORS, role checks on every protected route	Customer, farmer and admin routes are separated on the server, not only hidden in the menu
Data	MongoDB with the Mongoose ODM	19 collections with validation rules and indexes (section 4)
Services	Nodemailer (SMTP e-mail), an hourly scheduler, OpenStreetMap tiles and OSRM routes	Weekly stock template, low-stock alerts, "market open today" notices

2.2 Modules

Module	Main functions
Accounts	Register (customer, farmer wizard), one login page for every role, forgot / reset password, profile photo, family sharing
Markets and maps	Market list (opens on the markets open today), market page with the farmers there and whether each is at the market today, map of markets and stalls, driving route

Module	Main functions
Catalogue	Products with photos, categories, filters (price, category, market, day, rating, offers), grid or list view, search while typing, best sellers, product details with reviews
Pre-orders	Basket, checkout with a pickup date and slot per farmer, guest checkout, bookmarkable order page (/checkout/<order number>), modify and cancel before the cut-off
Farmer portal	Products and weekly stock template, inventory history, low-stock alerts, pickup windows, closed dates, "I cannot come today", orders (accept, decline, ready, complete), sales insights, review replies
Reviews	Verified reviews after pickup, "Did you receive your order?" question, farmer replies, moderation
Notifications	In-app and e-mail alerts for orders, restock reminders, new farmers, markets open today, announcements
Administration	Dashboard, users, markets, cities, categories, products, reviews, moderation queue, announcements, offer banner, messages by topic, newsletter, FAQs, reports, farmer rankings
AI assistant	Chat in English, Urdu and Roman Urdu: timings, top-rated farmers, best sellers, cheapest items, offers, farmer availability, order status and help

2.3 User interface design

- **Look:** one dark green (#173B2C) and one lime accent (#D4F06E) on a cream background, white cards and real photos (no illustrations); Fraunces for headings, Plus Jakarta Sans for text, Noto Nastaliq Urdu for Urdu.
- **Home page:** a full-width photo banner, then sections that take turns between dark green and light cream: categories, the offer banner, top picks, the video tour, top-rated farmers, markets and map, the farm-to-table story, an invitation for farmers and market organisers, customer reviews, why choose us and the FAQ.
- **Phones:** a bottom navigation bar on every page, dialogs open as bottom sheets, filters slide in from the side, tables turn into cards.
- **Accessibility:** keyboard access to every control, visible focus, alt text on every image, reduced motion respected where it matters, colour contrast checked.

2.4 Non-functional requirements

Requirement (SRS 1.7)	How it is met
Compatibility	Tested in Chromium; uses standard HTML, CSS and JavaScript supported by current Chrome, Edge, Firefox and Safari
Responsive design	Checked at 18 widths from 320 to 1920 px, no sideways scrolling (section 7 of the test report)
Security	Hashed passwords, httpOnly cookies, role checks on the server, rate limits, file-type checks on uploads
Performance	Server-side paging for tables, compressed responses, lazy-loaded pages, images sized for the screen (1024 px on phones, 1920 px on large screens)
Usability	Toast messages for every action, spinners on buttons while they work, clear empty states
Languages	English and Urdu on every public, customer and farmer page

3. Diagrams

3.1 Use cases

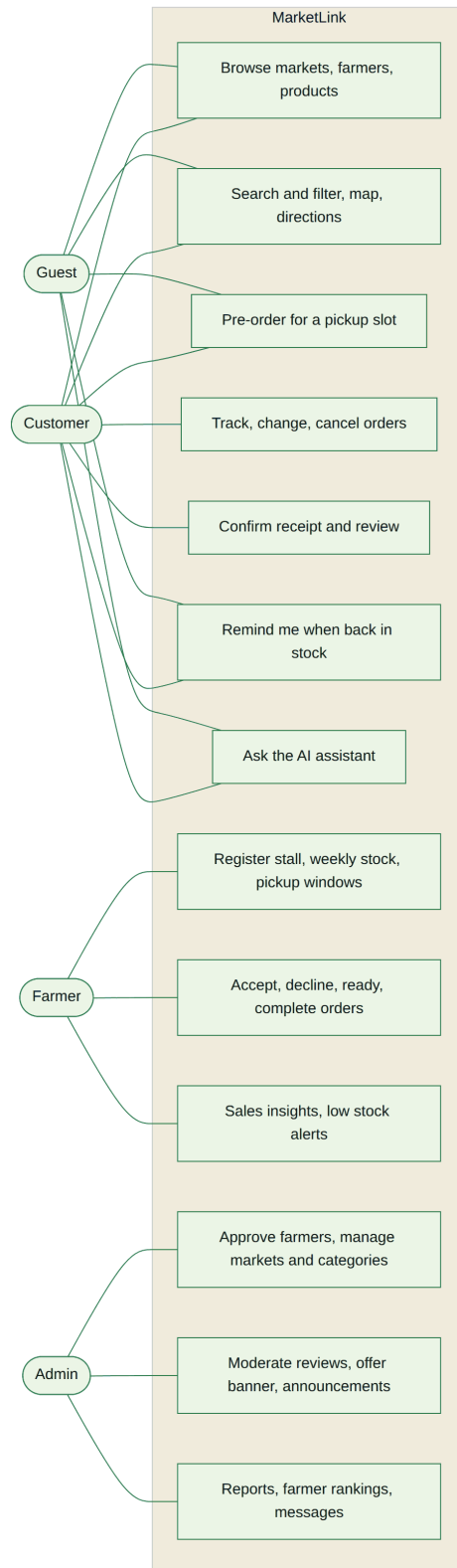


Figure 2 – Use cases by role

3.2 Data flow diagram – level 0 (context)

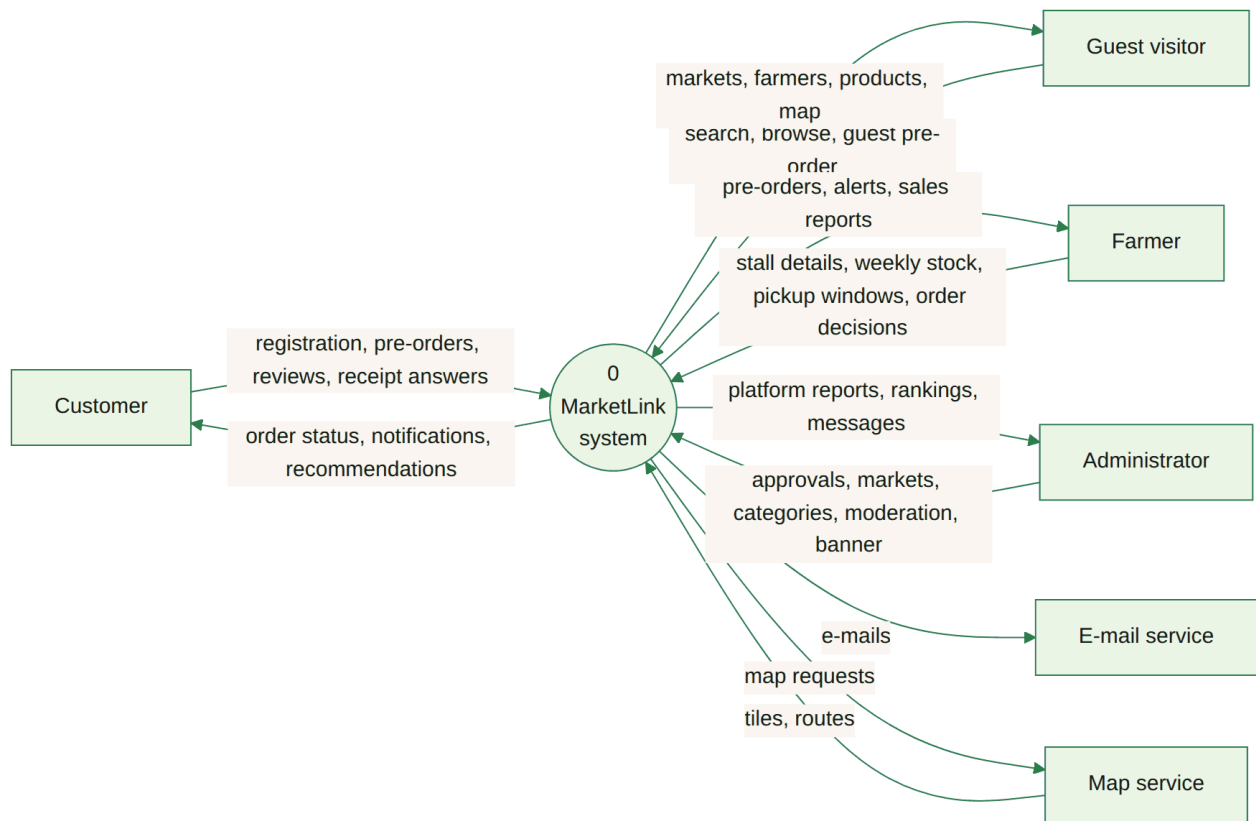


Figure 3 – Context diagram

3.3 Data flow diagram – level 1

The level-1 diagram is split in two to keep it readable.

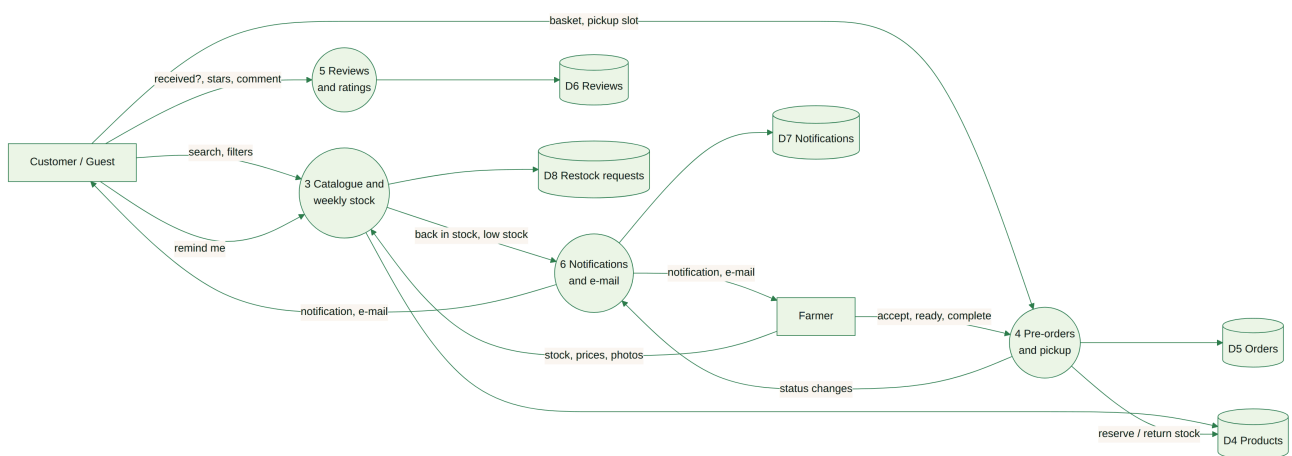


Figure 4 – DFD level 1: shopping, orders, reviews and notifications

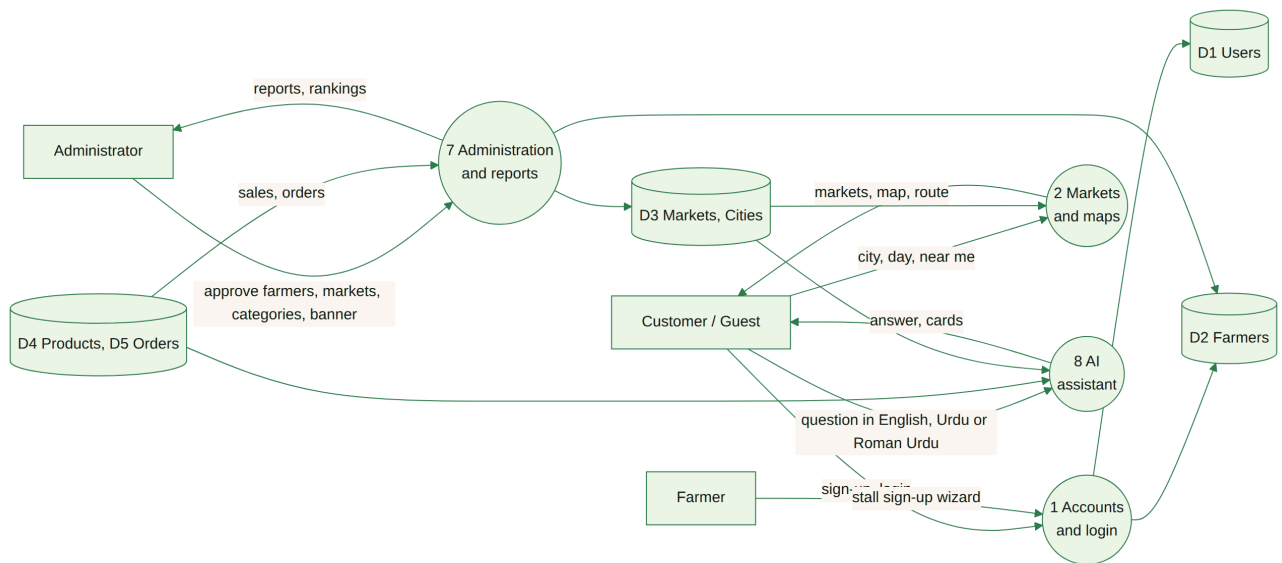


Figure 5 – DFD level 1: accounts, markets, administration and the assistant

3.4 Flowchart – customer pre-order

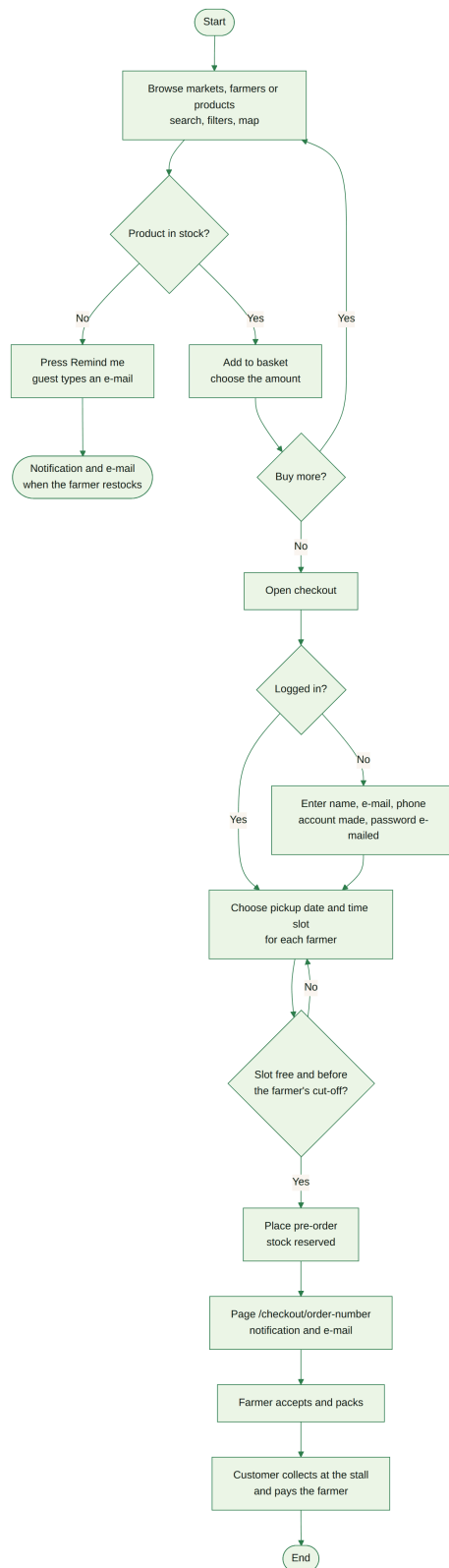


Figure 6 – Customer pre-order

3.5 Flowchart – farmer registration and approval

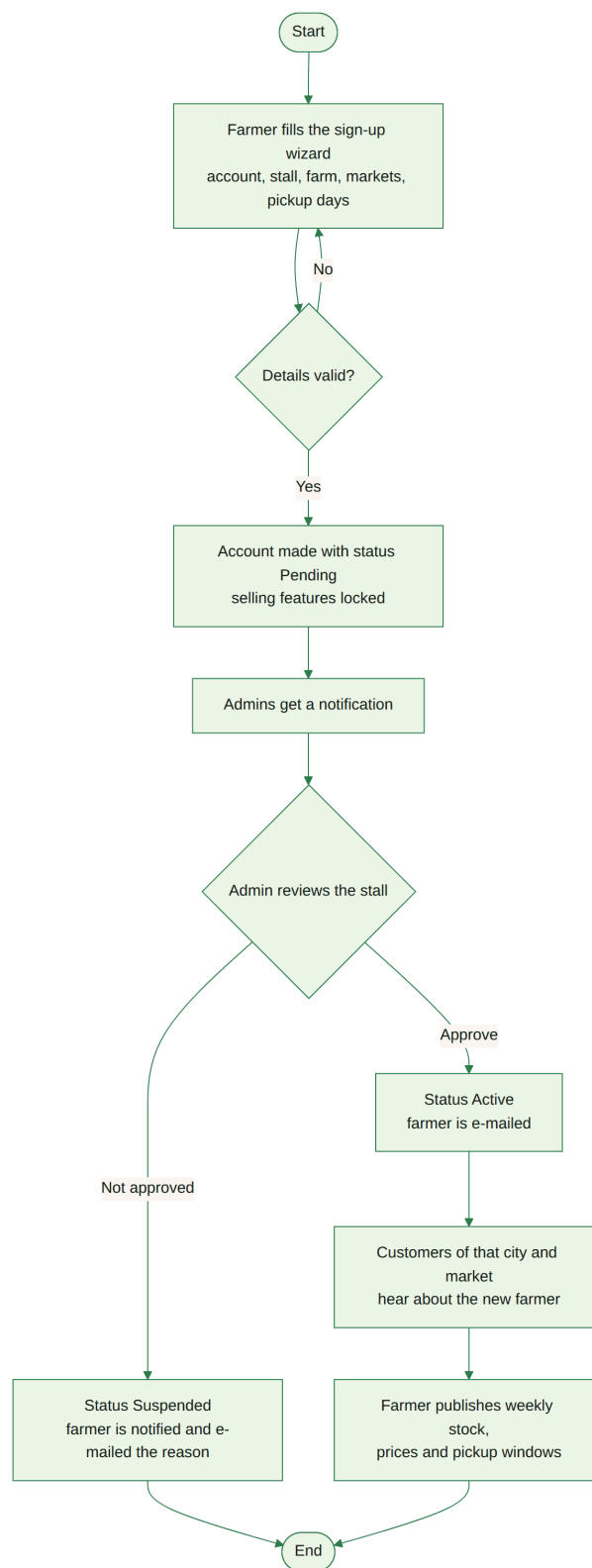


Figure 7 – Farmer registration and approval

3.6 Flowchart – farmer handles a pre-order

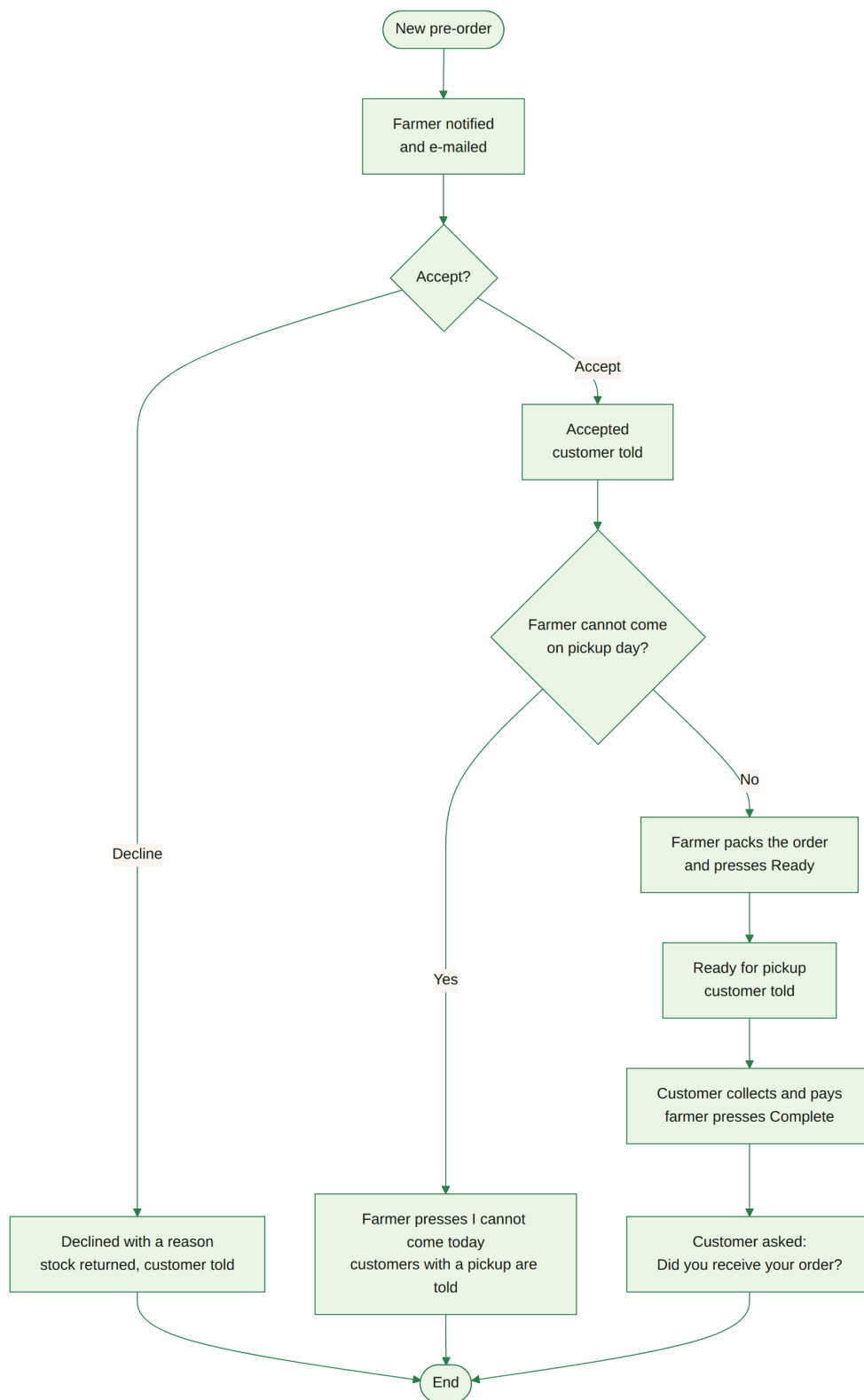


Figure 8 – Farmer order handling

3.7 Flowchart – receipt confirmation and review

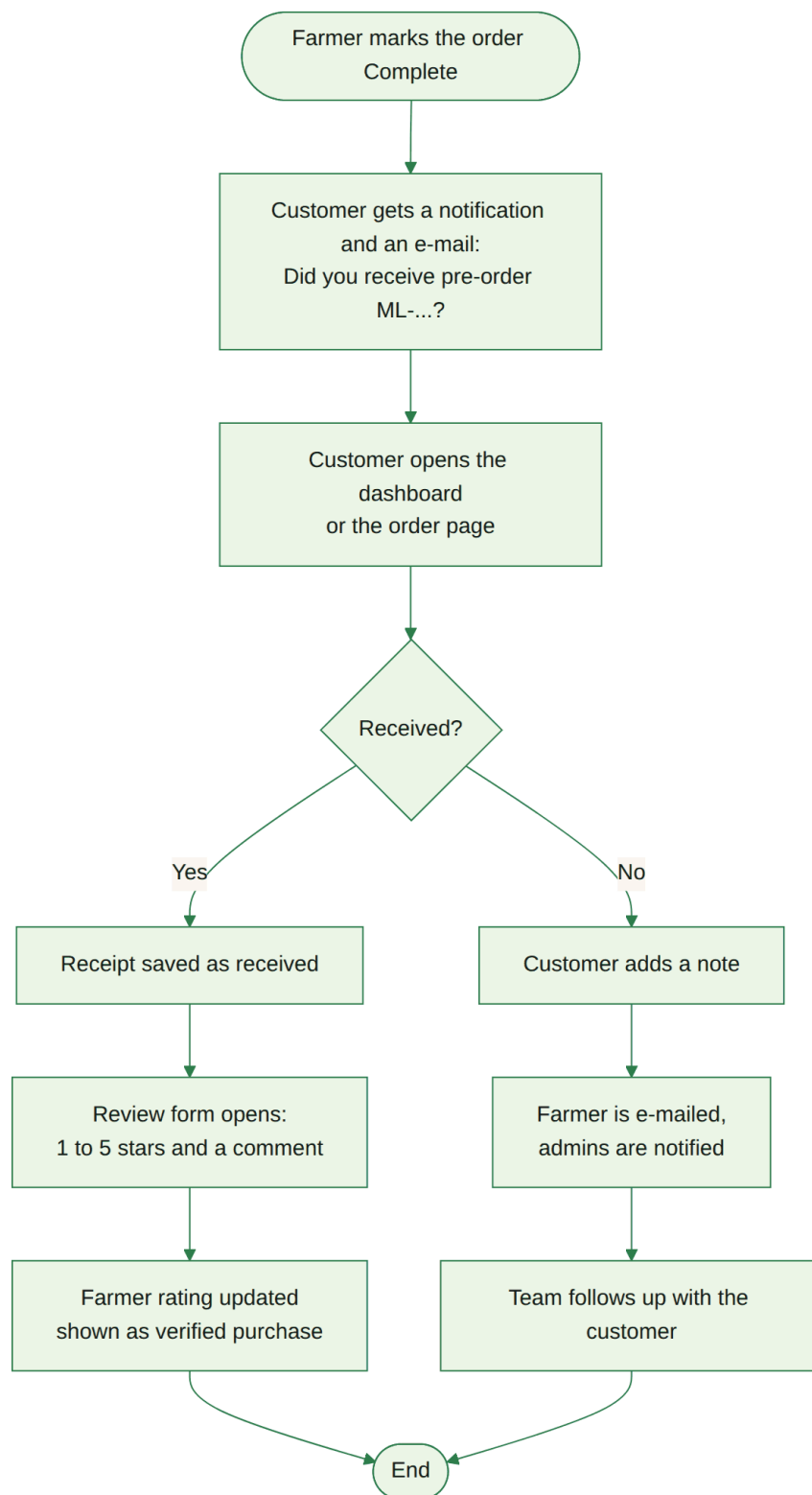


Figure 9 – Receipt confirmation and review

3.8 Flowchart – AI assistant

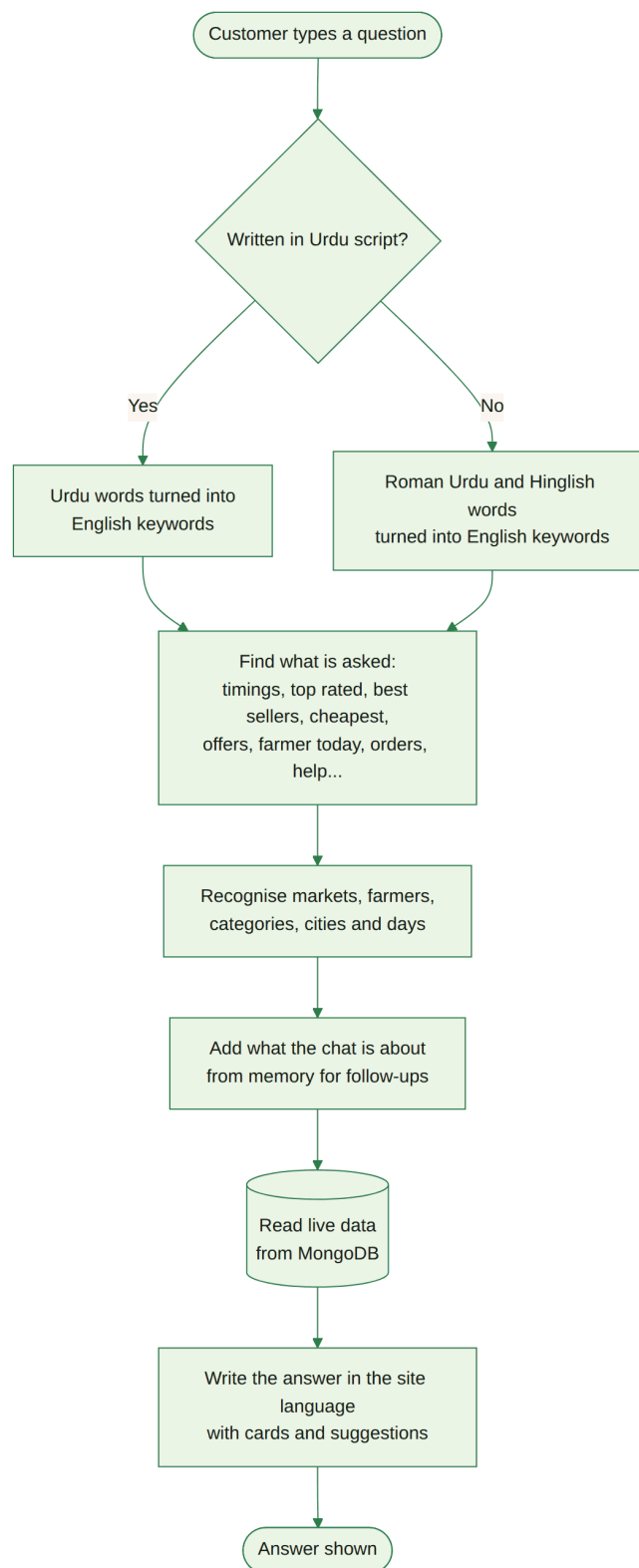


Figure 10 – How the assistant answers

3.9 Order states

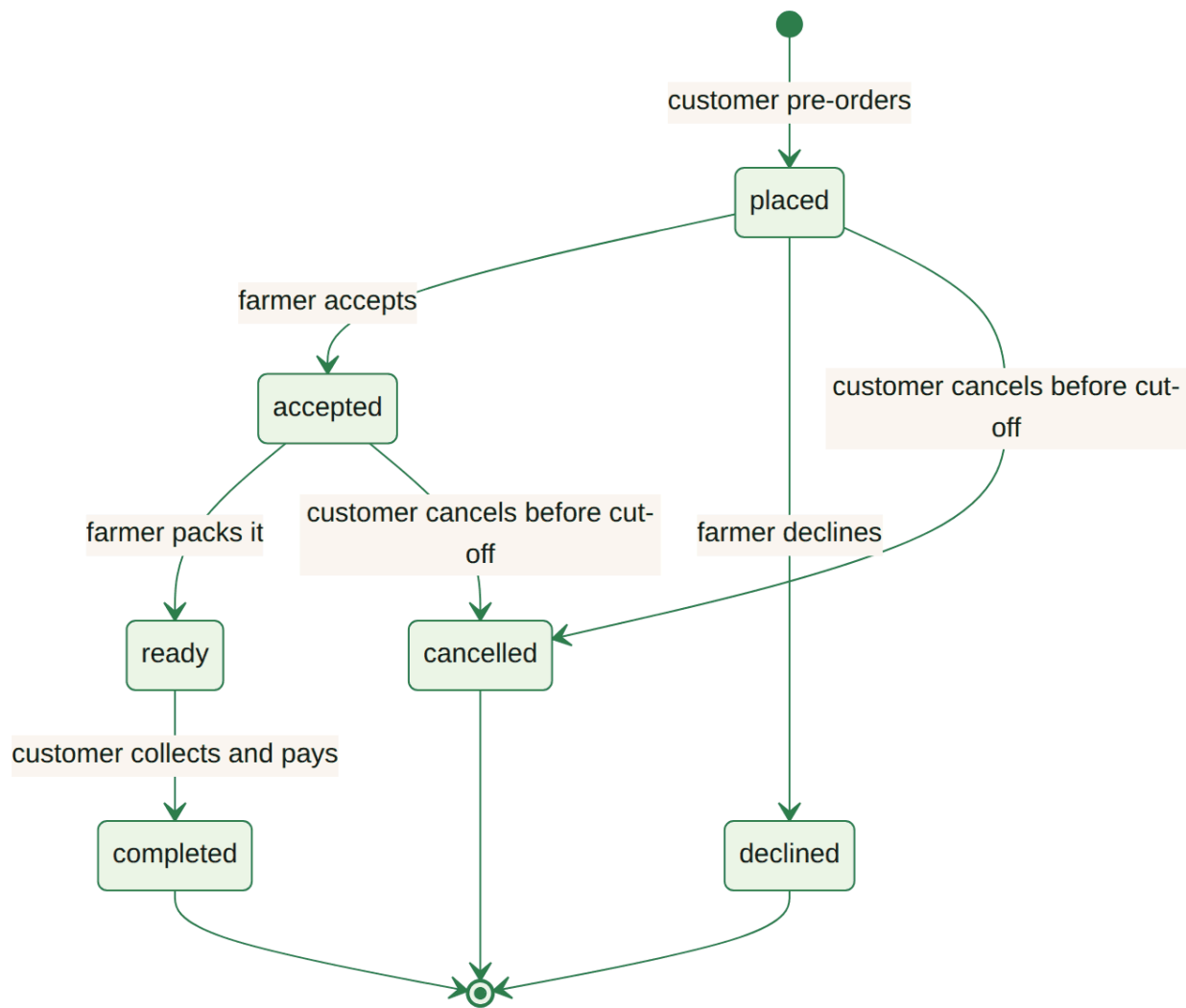


Figure 11 – Order status diagram

4. Database design

MongoDB stores the data in **19 collections**. The main entities and their relations are shown below; the full list of fields follows. Relations are stored as references (ObjectId) except the order items and pickup windows, which are stored inside their order or farmer document.

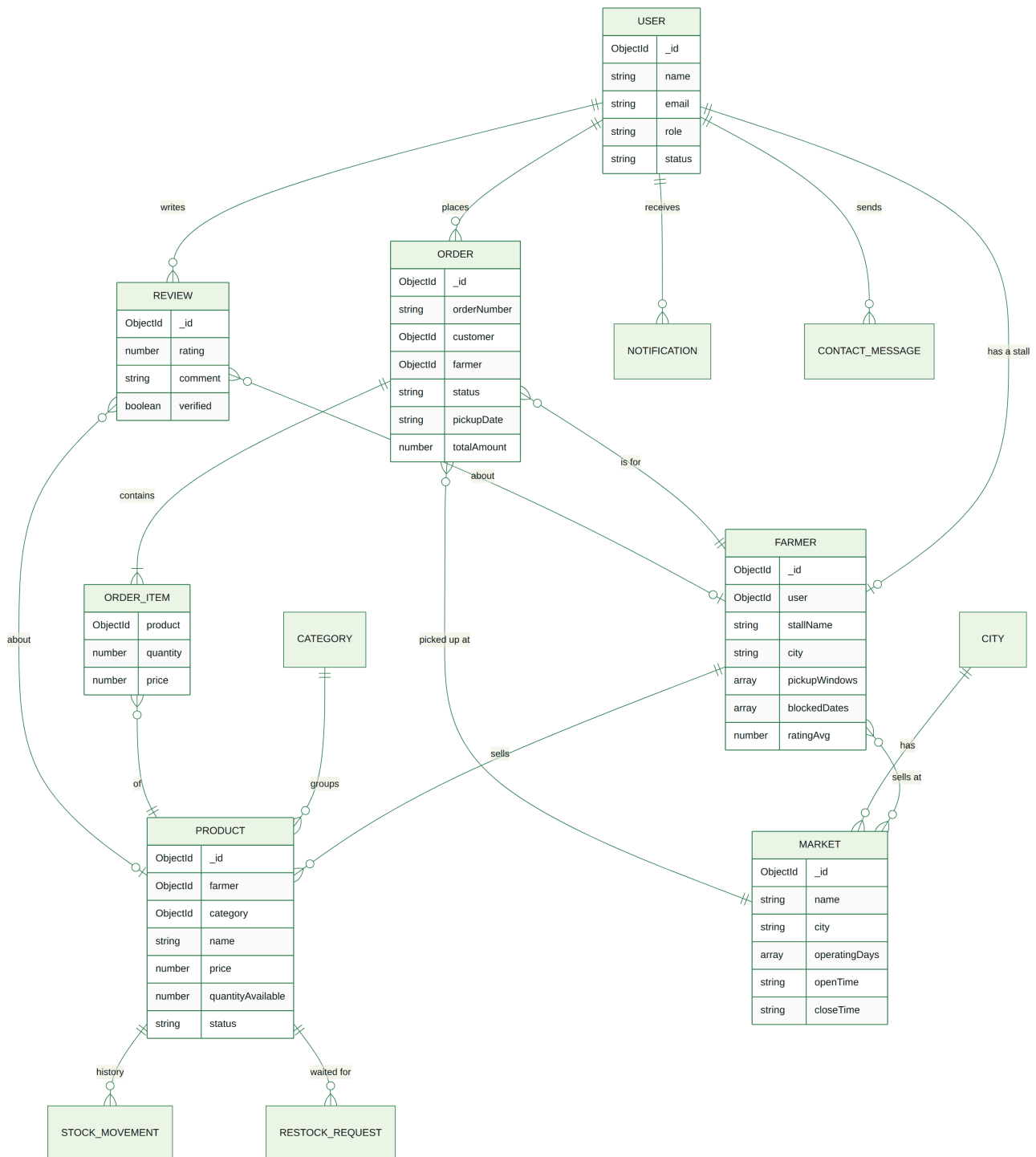


Figure 12 – Entity relationship diagram

4.1 Collections

Collection	Purpose
users	Every login: customers, farmers and administrators (role, status, favourites, family)
farmers	Stall profile of a farmer user: markets, days, pickup windows, closed dates, rating
markets	Farmers markets with address, days, hours and map position
cities	Master list of cities for markets and farmers
categories	Product categories (with Urdu name, photo and colour)
products	A farmer's product with price, unit, stock, status, photos and SEO fields
orders	Pre-orders with items, pickup slot, status history and receipt answer
reviews	Ratings and comments for farmers and products, with the farmer's reply
notifications	In-app messages for each user
stockmovements	History of every stock change (orders, restocks, template, corrections)
restockrequests	"Remind me when it is back" requests (customer or guest e-mail)
announcements	Platform notices, optionally for a season (months)
sitebanners	The home page offer banner the admin edits
contactmessages	Contact form messages with their topic
contentflags	Reports of inappropriate content for the moderation queue
faqs	Questions and answers (English and Urdu)
reports	Saved platform reports
subscribers	Newsletter subscribers
assistantchats	Saved chat history and memory of signed-in users

4.2 Fields of each collection

The tables below list every field with its type and the rules checked by the database layer.

Announcement (collection announcements)

Field	Type	Required	Rules / notes
title	String	Yes	max length 120
message	String	Yes	max length 1000
titleUr	String		max length 160
messageUr	String		max length 1200
audience	String		one of: all, customer, farmer; default "all"
months	Array		default []
link	String		max length 200
isActive	Boolean		default true

Field	Type	Required	Rules / notes
createdBy	ObjectId → User		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

AssistantChat (collection assistantchats)

Field	Type	Required	Rules / notes
user	ObjectId → User	Yes	unique
messages	Array of objects (from, text, cards, at)		
memory.marketId	String		
memory.marketName	String		
memory.farmerId	String		
memory.farmerName	String		
memory.productId	String		
memory.productName	String		
memory.day	Number		
memory.city	String		
memory.name	String		
memory.lastIntent	String		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: user 1 (unique)

Category (collection categories)

Field	Type	Required	Rules / notes
name	String	Yes	unique; max length 60
nameUr	String		max length 60
slug	String	Yes	unique
description	String		max length 300
icon	String		
image	String		
color	String		default "#E6F4DA"
sortOrder	Number		default 0

Field	Type	Required	Rules / notes
isActive	Boolean		default true
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: name 1 (unique); slug 1 (unique)

City (collection cities)

Field	Type	Required	Rules / notes
name	String	Yes	max length 60
slug	String	Yes	
province	String		max length 60; default ""
latitude	Number		min -90; max 90
longitude	Number		min -180; max 180
isActive	Boolean		default true
sortOrder	Number		default 0
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: name 1 (unique); slug 1 (unique)

ContactMessage (collection contactmessages)

Field	Type	Required	Rules / notes
name	String	Yes	max length 80
email	String	Yes	max length 120
subject	String		max length 150
topic	String		one of: market_request, market_complaint, farmer_complaint, order_help, selling, feedback, other
message	String	Yes	max length 2000
status	String		one of: new, read; default "new"
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

ContentFlag (collection contentFlags)

Field	Type	Required	Rules / notes
targetType	String	Yes	one of: review, product, farmer
review	ObjectId → Review		
product	ObjectId → Product		
farmer	ObjectId → Farmer		
reason	String	Yes	one of: spam, offensive, misleading, wrong_info, other, auto_language
note	String		max length 500
reporter	ObjectId → User		
status	String		one of: open, resolved, dismissed; default "open"
action	String		one of: none, removed, restored, suspended; default "none"
resolutionNote	String		max length 300
resolvedBy	ObjectId → User		
resolvedAt	Date		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: status 1, createdAt -1; targetType 1, review 1, product 1, farmer 1

Faq (collection faqs)

Field	Type	Required	Rules / notes
question	String	Yes	max length 200
answer	String	Yes	max length 1500
questionUr	String		max length 300
answerUr	String		max length 2500
group	String		one of: shopping, pickup, farmers, account; default "shopping"
order	Number		default 0
showOnHome	Boolean		default false
isActive	Boolean		default true
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: isActive 1, group 1, order 1

Farmer (collection farmers)

Field	Type	Required	Rules / notes
user	ObjectId → User	Yes	unique
stallName	String	Yes	max length 100
slug	String	Yes	unique
contactPerson	String	Yes	
phone	String	Yes	
email	String	Yes	
address	String	Yes	
city	String		default ""
bio	String		max length 1200
bioUr	String		max length 1500
tags	Array		
categories	Array		
logo	String		
coverImage	String		
coverCredit.author	String		
coverCredit.source	String		
coverCredit.license	String		
latitude	Number		min -90; max 90
longitude	Number		min -180; max 180
markets	Array		
operatingDays	Array		
pickupWindows	Array of objects (market, day, start, end)		
slotMinutes	Number		min 10; max 240; default 30
slotCapacity	Number		min 1; max 100; default 6
orderCutoffHours	Number		min 0; max 168; default 12
blockedDates	Array		
autoApplyTemplate	Boolean		default true
templateLastAppliedWeek	String		
isActive	Boolean		default false
ratingAvg	Number		default 0
ratingCount	Number		default 0
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically

Field	Type	Required	Rules / notes
updatedAt	Date		set automatically

Indexes: user 1 (unique); slug 1 (unique); isActive 1; markets 1; operatingDays 1

Market (collection markets)

Field	Type	Required	Rules / notes
name	String	Yes	max length 100
slug	String	Yes	unique
description	String		max length 1000
descriptionUr	String		max length 1200
address	String	Yes	
city	String		default ""
categories	Array		
latitude	Number	Yes	min -90; max 90
longitude	Number	Yes	min -180; max 180
mapProvider	String		one of: openstreetmap, google; default "openstreetmap"
mapLink	String		
operatingDays	Array		
openTime	String		default "07:00"
closeTime	String		default "13:00"
image	String		
imageCredit.author	String		
imageCredit.source	String		
imageCredit.license	String		
isActive	Boolean		default true
liveNoticeDate	String		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: slug 1 (unique); isActive 1, city 1; categories 1

Notification (collection notifications)

Field	Type	Required	Rules / notes
user	ObjectId → User	Yes	

Field	Type	Required	Rules / notes
type	String		one of: order, restock, stock, announcement, review, account, moderation, system; default "system"
title	String	Yes	
message	String	Yes	
link	String		
read	Boolean		default false
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: user 1, read 1, createdAt -1

Order (collection orders)

Field	Type	Required	Rules / notes
orderNumber	String	Yes	unique
customer	ObjectId → User	Yes	
farmer	ObjectId → Farmer	Yes	
market	ObjectId → Market	Yes	
items	Array of objects (product, name, nameUr, image, unit, price)		
totalAmount	Number	Yes	min 0
pickupDate	String	Yes	
pickupSlot.start	String	Yes	
pickupSlot.end	String	Yes	
pickupAt	Date	Yes	
cutoffAt	Date	Yes	
status	String		one of: placed, accepted, ready, completed, declined, cancelled; default "placed"
statusHistory	Array of objects (status, at, by, note)		
customerNote	String		max length 500
placedBy	String		one of: customer, admin; default "customer"
farmerNote	String		max length 500
paymentMethod	String		default "pay_at_pickup"
completedAt	Date		
receipt.status	String		one of: received, not_received
receipt.note	String		max length 500

Field	Type	Required	Rules / notes
receipt.at	Date		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: orderNumber 1 (unique); customer 1, createdAt -1; farmer 1, status 1, pickupDate 1; market 1

Product (collection products)

Field	Type	Required	Rules / notes
farmer	ObjectId → Farmer	Yes	
name	String	Yes	max length 100
nameUr	String		max length 100
slug	String		
category	ObjectId → Category	Yes	
price	Number	Yes	min 0
compareAtPrice	Number		min 0
unit	String		one of: kg, g, lb, dozen, piece, bunch, litre, pack, jar, loaf, box; default "kg"
quantityAvailable	Number		min 0; default 0
templateQuantity	Number		min 0; default 0
lowStockThreshold	Number		min 0; max 100000; default 5
lowStockAlertedAt	Date		
soldOutAlertedAt	Date		
description	String		max length 1500
descriptionUr	String		max length 1800
metaTitle	String		max length 70
metaDescription	String		max length 170
keywords	Array		default []
aiSchema.summary	String		max length 300
aiSchema.season	String		max length 80
aiSchema.storage	String		max length 200
aiSchema.uses	String		max length 200
aiSchema.usesUr	String		max length 250
aiSchema.storageUr	String		max length 250
aiSchema.source	String		one of: claude, builtin, farmer
aiSchema.generatedAt	Date		

Field	Type	Required	Rules / notes
image	String		
gallery	Array of objects (url, credit.author, credit.source, credit.license)		
imageCredit.author	String		
imageCredit.source	String		
imageCredit.license	String		
status	String		one of: available, sold_out, unavailable; default "available"
isRemoved	Boolean		default false
removedReason	String		
deletedByFarmer	Boolean		default false
markets	Array		
days	Array		
farmerActive	Boolean		default true
ratingAvg	Number		default 0
ratingCount	Number		default 0
totalSold	Number		default 0
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: slug 1; farmer 1, isRemoved 1; category 1, price 1; markets 1; days 1; keywords 1

Report (collection reports)

Field	Type	Required	Rules / notes
generatedBy	ObjectId → User	Yes	
reportType	String	Yes	one of: platform_overview, orders_summary, revenue_by_market, top_farmers, sales_by_category, customer_activity, inventory_status, city_overview, reviews_moderation
title	String		
from	Date		
to	Date		
data	Mixed		
_id	ObjectId	Yes	primary key
generatedAt	Date		

RestockRequest (collection restockrequests)

Field	Type	Required	Rules / notes
product	ObjectId → Product	Yes	
user	ObjectId → User		
email	String	Yes	max length 120
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: product 1, email 1 (unique)

Review (collection reviews)

Field	Type	Required	Rules / notes
type	String	Yes	one of: product, farmer
product	ObjectId → Product		
farmer	ObjectId → Farmer	Yes	
customer	ObjectId → User	Yes	
order	ObjectId → Order		
verified	Boolean		default false
rating	Number	Yes	min 1; max 5
comment	String		max length 1000
response.text	String		max length 1000
response.at	Date		
isRemoved	Boolean		default false
removedReason	String		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: product 1, isRemoved 1; farmer 1, isRemoved 1; customer 1, order 1

SiteBanner (collection sitebanners)

Field	Type	Required	Rules / notes
key	String	Yes	unique; max length 40
isActive	Boolean		default true
percent	Number		min 0; max 90; default 30
autoPercent	Boolean		default false

Field	Type	Required	Rules / notes
tag	String		max length 40
tagUr	String		max length 60
title	String		max length 90
titleUr	String		max length 120
text	String		max length 220
textUr	String		max length 300
buttonLabel	String		max length 30
buttonLabelUr	String		max length 40
link	String		max length 200
image	String		
updatedBy	ObjectId → User		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: key 1 (unique)

StockMovement (collection stockmovements)

Field	Type	Required	Rules / notes
farmer	ObjectId → Farmer	Yes	
product	ObjectId → Product	Yes	
productName	String	Yes	
productNameUr	String		
unit	String		
change	Number	Yes	
quantityAfter	Number	Yes	min 0
type	String	Yes	one of: initial, restock, adjustment, waste, stall_sale, correction, template, order_reserved, order_released, order_changed
reason	String		max length 200
order	ObjectId → Order		
orderNumber	String		
by	String		one of: farmer, customer, admin, system; default "system"
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically

Field	Type	Required	Rules / notes
updatedAt	Date		set automatically

Indexes: farmer 1, createdAt -1; product 1, createdAt -1

Subscriber (collection subscribers)

Field	Type	Required	Rules / notes
email	String	Yes	max length 120
name	String		max length 80
source	String		one of: home, footer, checkout, admin; default "footer"
status	String		one of: subscribed, unsubscribed; default "subscribed"
token	String		
unsubscribedAt	Date		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: email 1 (unique); token 1

User (collection users)

Field	Type	Required	Rules / notes
name	String	Yes	max length 80
email	String	Yes	unique
password	String	Yes	
role	String		one of: customer, farmer, admin; default "customer"
phone	String		
address	String		max length 250
city	String		max length 60
status	String		one of: active, pending, suspended, inactive; default "active"
avatar	String		
favoriteFarmers	Array		
favoriteProducts	Array		
savedMarkets	Array		
household	ObjectId → User		
lastLoginAt	Date		
termsAcceptedAt	Date		
termsVersion	String		

Field	Type	Required	Rules / notes
resetPasswordHash	String		
resetPasswordExpires	Date		
_id	ObjectId	Yes	primary key
createdAt	Date		set automatically
updatedAt	Date		set automatically

Indexes: email 1 (unique); role 1, status 1

4.3 Database scripts

MongoDB does not use SQL, so the SRS's `.sql` files are replaced by:

- `database/marketlink-schema.mongoddb.js` – creates the collections with their validation rules and indexes (run with `mongosh`).
- `database/sample-data/` – the demo data as JSON files that can be imported with MongoDB Compass or `mongoimport`.
- `npm run seed` – fills a fresh database with the same demo data.

5. Test data used

The demo data is created by `npm run seed` and is the data used for every test in the test report.

Record counts after `npm run seed`

Data	Records	Details
Users	21	1 administrator; 12 farmers (10 approved, 1 waiting for approval, 1 suspended); 8 customers (7 active, 1 deactivated)
Cities	9	Karachi, Lahore, Islamabad, Rawalpindi, Hyderabad, Faisalabad, Multan, Peshawar, Quetta
Markets	8	6 in Karachi, 1 in Lahore, 1 in Islamabad (table below)
Categories	8	Vegetables, Fruits, Dairy & Eggs, Baked Goods, Herbs & Greens, Honey & Preserves, Grains & Pulses, Flowers & Plants
Products	62	15 Fruits, 15 Vegetables, 7 Baked Goods, 6 Flowers & Plants, 6 Dairy & Eggs, 5 Honey & Preserves, 4 Herbs & Greens, 4 Grains & Pulses; 60 in stock, 2 sold out; 8 on offer; prices Rs 50 to Rs 3,200
Pre-orders	435	Pickup dates 1 August to 2 October 2026: 371 completed, 24 cancelled, 32 declined, and upcoming ones: 3 placed, 3 accepted, 2 ready
Reviews	341	188 farmer reviews and 153 product reviews, most with a verified purchase; some with farmer replies
Stock movements	81	Restocks, weekly template runs, corrections and order reservations
Notifications	11	Order updates, alerts and announcements for the demo accounts
Announcements	5	Including seasonal notices shown only in their months
FAQs	17	English and Urdu, some shown on the home page
Content flags	3	Reports waiting in the moderation queue
Contact messages	2	Example messages for the admin inbox
Newsletter subscribers	6	
Saved reports	1	An example platform report

Markets

Market	City	Days	Hours
Clifton Seaview Farmers Market	Karachi	Sunday, Saturday	07:00 to 13:00
DHA Phase 6 Weekend Market	Karachi	Sunday, Friday	08:00 to 14:00
Gulshan Sunday Kisan Bazaar	Karachi	Sunday, Wednesday	07:00 to 15:00
North Nazimabad Evening Green Market	Karachi	Wednesday, Saturday	16:00 to 21:00
Bahadurabad Fresh Friday Market	Karachi	Tuesday, Friday	16:00 to 21:00
Saddar Heritage Farmers Market	Karachi	Monday, Thursday, Saturday	06:00 to 12:00
Model Town Kisan Market	Lahore	Sunday, Thursday	07:00 to 13:00

Market	City	Days	Hours
F-9 Park Weekend Market	Islamabad	Sunday, Saturday	08:00 to 14:00

Farmers

Stall	City	Status	Rating (reviews)
Malir Green Fields	Karachi	Approved	4.55 (56)
Gadap Orchard Co.	Karachi	Approved	4.29 (41)
Thatta Dairy Collective	Karachi	Approved	4.28 (39)
Karachi Artisan Bakehouse	Karachi	Approved	4.41 (39)
Hyderabad Herb Garden	Karachi	Approved	4.25 (32)
Kirthar Honey & Preserves	Karachi	Approved	4.41 (17)
Indus Grains & Pulses	Karachi	Approved	4.49 (43)
Bloom & Bough Nursery	Karachi	Approved	4.52 (25)
Ravi Organic Farm	Lahore	Approved	4.25 (20)
Margalla Hills Farm	Islamabad	Approved	4.36 (28)
Sunny Acres Poultry	Karachi	Waiting for approval	-
Old Town Goat Dairy	Karachi	Suspended	-

Example inputs used in the tests

Input	Values used
New accounts	Customer "New User" (new@x.com, +92 300 1234567, password Passw0rd); a sign-up without accepting the terms (noterms@x.com, refused); a farmer through the sign-up wizard; the browser test types the phone as +92 321 5551234 and checks it is saved as +923215551234
Invalid inputs	Empty required fields, wrong e-mail format, short password, phone with letters, price below zero, percent over 90, an outside web link
Pre-orders	1 to 3 items from one or two farmers, the next free pickup slot, guest checkout with a new e-mail
Search words	mango, tomato, honey, "hon" (partial), Urdu and Roman Urdu words such as "tamatar"
Assistant questions	"which farmer is highly rated", "sab se acha kisan kaunsa hai?", "tamatar kahan milega", "sab se sasti sabzi", Urdu "سب سے اچھی ریٹنگ والا کسان کون ہے؟"

Special cases in the data, so every screen can be tried:

- A farmer waiting for approval (Sunny Acres Poultry) and a suspended farmer (Old Town Goat Dairy).
- A deactivated customer, and a family account (Omar Khan shares Ayesha Khan's household).
- Products on offer (a higher "before" price), sold-out and low-stock products.
- Pre-orders in every status, including upcoming ones for today and the coming days, and one completed pickup the demo customer has not yet confirmed (so "Did you receive your order?" can be tried).
- A closed date for Bloom & Bough Nursery next Friday, reviews with farmer replies, reported content in the moderation queue, seasonal announcements and newsletter subscribers.

6. Installation instructions

Requirements: Node.js 20 or newer, and MongoDB (MongoDB Community Server on the computer, or a free MongoDB Atlas cluster).

1. Open a terminal in the `MarketLink` folder.
2. Install everything: `npm run install:all`
3. Copy `server/.env.example` to `server/.env` and set `MONGO_URI` (for example `mongodb://127.0.0.1:27017/marketlink`) and a long random `JWT_SECRET`.
4. (Optional) Create the collections and indexes: run `database/marketlink-schema.mongoose.js` with `mongosh`.
5. Load the demo data: `npm run seed` (this deletes existing MarketLink data).
6. Start the app for development: `npm run dev`, then open `http://localhost:5173`
7. Or build and run one server: `npm run build` and `npm start`, then open `http://localhost:5000`

E-mail: without SMTP settings the e-mails are printed in the server window. To send real e-mails, put the SMTP details (for example a Gmail address and app password) in `server/.env` and run `npm run mail:test`.

Urdu: press اردو in the top bar, or open any page with `?lang=ur`.

Ready-to-run copy: the "Complete" zip already contains the built app and the server's packages, so only MongoDB, `server/.env` and `npm run seed` / `npm start` are needed (see `HOW-TO-RUN.txt`).

7. User credentials

All accounts are created by `npm run seed`. Everyone logs in on the same page, `/login`.

Role	E-mail	Password
Administrator	admin@marketlink.com	Admin@123
Farmer (approved) – Malir Green Fields	farmer@marketlink.com	Farmer@123
Farmer (waiting for approval) – Sunny Acres Poultry	pending.farmer@marketlink.com	Farmer@123
Farmer (suspended, cannot log in) – Old Town Goat Dairy	suspended.farmer@marketlink.com	Farmer@123
Other approved farmers	orchard@, dairy@, bakery@, herbs@, honey@, grains@, nursery@, lahore.farm@, islamabad.farm@ marketlink.com	Farmer@123
Customer – Ayesha Khan	customer@marketlink.com	Customer@123
Customer – Omar Khan (Ayesha's family member)	omar@marketlink.com	Customer@123
Other customers	bilal@, sara@, usman@, fatima@, hamza@ marketlink.com	Customer@123
Customer (deactivated, cannot log in)	inactive.customer@marketlink.com	Customer@123

8. SRS functional requirements and where to find them

SRS requirement	Where it is in MarketLink
Customer registration and login with name, contact number, e-mail and address	/register , /login ; phone box with country flag
Favourite farmers and products	Heart buttons; Account → Favourites
Family account sharing (optional)	Account → Profile → Family
Browse markets by location and day, farmers at each market	/markets (open today first), market page with each farmer's status today
Farmer profile: stall, location, days, weekly stock	/farmers/<name>
Map with markers and directions	/map , market and farmer pages (OpenStreetMap, OSRM route, Google Maps link)
Categories and filters for price, category, market and day	/products filters, /categories
Product details: price, unit, quantity, farmer	/products/<name>
Basket and pre-order against stock, pickup date and slot	Basket, /checkout
Order status, cancel or modify before cut-off	Account → Orders → order page
Order history, reorder, favourites, restock alerts	Account → Orders, "Order again", Remind me
Preferred market and route-friendly pickup details	Account → Favourites → Saved markets, order page directions
AI assistant	Chat button on every page
Reviews and ratings after completion, visible before ordering	Receipt question, review form, product and farmer pages
Farmer registration details and profile (markets, days, windows, map pin)	/register/farmer wizard, Farmer → Profile, Pickup
Products: add, edit, view, delete; weekly template; sold out / unavailable	Farmer → Products, Inventory
Pre-orders: accept, decline, ready; cut-off and slots	Farmer → Orders, Pickup
Sales history, best sellers, total / pending orders, revenue	Farmer → Dashboard, Sales
Reply to reviews	Farmer → Reviews
Admin dashboard with platform totals	/admin
Approve or suspend farmers; activate or deactivate customers	Admin → Farmers, Customers
Manage markets with map position	Admin → Markets
Remove inappropriate listings or reviews	Admin → Moderation, Products, Reviews
Reports: orders, revenue by market, most active farmers	Admin → Reports, farmer rankings
Categories and announcements	Admin → Categories, Announcements (plus the offer banner)
Role-based access	Server checks the role on every protected request
Search, sort, filter with map results	Shop, farmers, markets, map pages
Responsive design	Every page, phone bottom bar

SRS requirement	Where it is in MarketLink
E-mail or in-app notifications	Bell menu, Notifications page, e-mails
About us, contact us with map	/about , /contact

9. Features added beyond the SRS

Urdu language with right-to-left layout; guest checkout; the "Did you receive your order?" check before reviews; restock reminders for guests and customers; "I cannot come today" for farmers and the live "at the market today" badge; notices when a market opens today or a new farmer joins; best sellers (top 5 / 10 / 20 per market); admin farmer rankings; an admin-edited offer banner; low-stock alerts and stock history; AI-written product descriptions; SEO (search-engine friendly pages, sitemap, structured data); contact topics (for example "Request a new market"); a newsletter; FAQs.

10. Assumptions

- Payment is made in person at pickup; there is no payment gateway (SRS 1.5).
- Pickup only; no delivery (SRS 1.5).
- Farmer identity and organic or food-safety claims are not verified; tags such as "Pesticide-free" are the farmer's own description (SRS 1.5).
- All markets use one time zone (Asia/Karachi by default) for pickup slots, cut-off times and "today".
- The e-mail address is the login name (the SRS example table's `username`).
- The project uses MongoDB, so the database is described with a `mongosh` script and JSON sample data instead of `.sql` files.
- OpenStreetMap is used instead of the paid Google Maps API; Google Maps links are offered for directions.
- Stock photos come from the Open Images Dataset under the CC BY 2.0 licence; their photographers are listed on the website's **Photo credits** page.

11. AI tools used (SRS requirement)

Tool	What it was used for
Claude Code (Anthropic)	Coding assistant during development, tests, and the first draft of this report and the test report
Canva	Logo design
EDSR super-resolution model (OpenCV)	Enlarging the 1024 px banner photos to 1920 px for large screens
<i>(add any other tools your team used)</i>	